

I. Voice Synthesis & Conversational Data Sizing Estimate

Gemma 4B / SLM Voice Fine-Tuning Estimate

To achieve hyper-realistic, human-sounding voice interaction for HyrLink, training must be split into two distinct layers: **Text-Level Conversational Nuance (SLM)** and **Audio Engine Training (TTS/STT Models)**.

12M - 15M SLM TEXT INSTRUCTION PAIRS (3X CURRENT 5M)	2.5M - 3.5M STT DISFLUENCY CLEANUP PAIRS	3.0M - 4.5M TTS PROSODY & SSML MARKUP PAIRS
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Breakdown of SLM Record Growth (Expanding 5M to 15M):

- STT Processing & Cleanup:** Requires ~3M raw ASR vs. cleaned text pairs (removing "um/uh", false starts, run-on sentences while preserving core intent).
- TTS Prosody Alignment:** Requires ~4M plain text vs. prosody-annotated pairs (adding SSML tags, pause durations <break time="300ms"/>, emotional cues [chuckle], [thoughtful pause], emphasis, and phonetic spelling).
- Candidate Proxy Chatbot Nuance:** Requires ~3M multi-turn conversational dialogue turns with speech-like backchanneling ("mhm", "I see", turn-taking markers).

II. Feature-Specific Dataset Directory

Feature 1: Resume Rewriting (Narrative Format)

callmeech / resume-score-details

TYPE: RESUME TO NARRATIVE MAPPING

Description: 1,031 paired resumes and JDs generated with GPT-4o containing rich micro/macro justification narratives and structured skill extractions.

Use Case: Fine-tunes the SLM to transform structured/bulleted experience blocks into coherent career story paragraphs.

<https://huggingface.co/datasets/callmeech/resume-score-details>

WikiBio Dataset (HuggingFace)

TYPE: STRUCTURED DATA TO NARRATIVE TEXT

Description: 728,321 articles mapping Wikipedia infobox tables (facts, bullet points, dates) directly to natural introductory bio paragraphs.

Use Case: Teaches the model how to convert raw, tabular career facts into fluid narrative biographies without omitting key metrics.

https://huggingface.co/datasets/wiki_bio

Feature 2: Candidate Deep Dives (Questionnaire)

AnthropicInterviewer Dataset

TYPE: QUALITATIVE INTERVIEW PROBING

Description: 1,250 qualitative, semi-structured interviews conducted by AI with human interviewees.

Use Case: Trains the SLM on how to ask probing follow-up questions to turn candidate raw, unpolished inputs into polished storylines.

<https://huggingface.co/datasets/Anthropic/AnthropicInterviewer>

Inquisitive Question Generation Dataset

TYPE: DEEP PROBING QUESTION PAIRINGS

Description: 19,000 text spans paired with elicitive questions designed to evoke deeper explanations and underlying motivations.

Use Case: Enables HyrLink to generate candidate-specific deep-dive questions based on vague initial answers.

<https://huggingface.co/datasets/inquisitive>

Feature 3: Skill Validation & Testing

SkillSpan & TechWolf Skill Benchmarks

TYPE: SKILL SPAN EXTRACTION & VALIDATION

Description: ESCO-mapped skill boundaries across technical and non-technical domains.

Use Case: Teaches the SLM to isolate specific skill claims and construct roleplay scenario tests to score actual competence.

<https://huggingface.co/datasets/jjzha/skillspan>

SWE-bench / CodeXGlue

TYPE: COMPLEX TECHNICAL SKILL VALIDATION

Description: Real-world software engineering issues, code solutions, and automated execution evaluations.

Use Case: Evaluates candidate technical submissions against ground-truth problem solutions to yield numeric validation scores.

<https://huggingface.co/datasets/princeton-nlp/SWE-bench>

Feature 4: Role-Fit Scoring

Candidate Job Role & Skill Gap Dataset

TYPE: WEIGHTED ROLE-FIT & GAP ANALYSIS

Description: Candidate profiles paired with target job descriptions, complete with explicit skill gap flags and match scores.

Use Case: Trains the model to compute weighted macro/micro suitability scores and write justification rationale paragraphs.

<https://www.kaggle.com/datasets/ckshetty/candidate-job-role-dataset>

AI Recruitment Pipeline Dataset

TYPE: RESUME-TO-JD MATCH & OUTCOME SCORING

Description: Recruitment pipeline data linking candidate resumes, test scores, and match ratings to final hiring outcomes.

Use Case: Calibrates candidate role-fit scoring against historical recruitment decision outcomes.

<https://www.kaggle.com/datasets/yaswanthkumary/ai-recruitment-pipeline-dataset>

Feature 5: Pro Chatbot (AI Proxy of Candidate)

Persona-Chat & Persona-2.0

TYPE: TRAIT-GROUNDED MULTI-TURN CHATBOT

Description: Multi-turn dialogues where agents converse while adhering strictly to assigned background facts, tone, and preferences.

Use Case: Ensures the Pro Chatbot maintains candidate voice, personality, and career preferences during recruiter interviews.

https://huggingface.co/datasets/facebook/persona_chat

Feature 6: Recruiter Job Postings & Candidate Pairing

Candidate Recommendation Model Datasets

TYPE: CANDIDATE POOL MATCHMAKING

Description: GitHub repositories and benchmark datasets mapping recruiter job requisitions to candidate pools.

Use Case: Fine-tunes the SLM to select the top candidates from a database and synthesize comparative selection rationales.

<https://github.com/Srikar-R/Candidate-Recommendation-Model>

Feature 7: Voice Synthesization (TTS Alignment & Prosody)

DisfluencySpeech (Single-Speaker Conversational TTS Dataset)

TYPE: PARALANGUAGE & PROSODY SPEECH DATASET

Description: Studio-quality conversational English speech dataset with annotated paralinguistic (non-speech events, false starts, discourse markers).

Use Case: Teaches the SLM to predictively insert paralinguistic markers (sighs, laughter, pauses) into text responses for TTS engines.

<https://huggingface.co/datasets/amaai-lab/DisfluencySpeech>

Expresso / SeamlessExpressive (Meta AI)

TYPE: EXPRESSIVE SPEECH PROSODY CORPUS

Description: High-quality speech dataset annotated with fine-grained prosodic features, pitch contours, speech rate, and emotion styles.

Use Case: Trains the model to output SSML prosody tags (<prosody pitch="..." rate="...">) to control downstream TTS style.

<https://huggingface.co/facebook/seamless-expressive>

LibriTTS-R (Clean Prosody Corpus)

TYPE: RESTORED SPEECH & PHONETIC ALIGNMENT

Description: 585 hours of clean, restored audio paired with normalized text, phonetic transcriptions, and duration tags.

Use Case: Trains the SLM on phonetic spelling hints and text normalization for crisp, natural TTS pronunciation.

https://huggingface.co/datasets/parler-tts/libritts_r_filtered

Feature 8: Speech-to-Text (STT) Processing & Cleanup

ami-disfluency (HuggingFace)

TYPE: DISFLUENCY & SPONTANEOUS SPEECH ANNOTATIONS

Description: 35,000+ annotated disfluency segments (false starts, stutters, filler words, laughter) in conversational meeting audio.

Use Case: Trains the SLM to clean up unedited spoken transcriptions from raw STT engines into clear, grammatical prose.

<https://huggingface.co/datasets/hhoangphuoc/ami-disfluency>

Switchboard Disfluency Transcripts

TYPE: RAW VS. CLEAN CONVERSATIONAL TEXT PAIRS

Description: Conversational telephone transcripts paired with explicit disfluency edit tags (restarts, filled pauses, discourse markers).

Use Case: Fine-tunes the SLM on intent-preserving disfluency removal—stripping filler words while preserving key context.

https://huggingface.co/datasets/switchboard_dps